

Name: Sam Shema

Class: Machine Learning

Machine Learning Project

Title:

Can a Computer Predict Heart Disease? A Comparison of Three Machine Learning Models

Abstract:

This project explores whether machine learning can accurately predict the presence of heart disease using patient medical data. Using a dataset of 1,025 patients from Kaggle (<https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>), which includes features such as age, cholesterol level, and blood pressure, I implemented three models: *Logistic Regression*, *Decision Trees*, and *Neural Networks*. The goal is to compare their performance and determine which model is most effective and practical for real-world medical use. Results showed that while more complex models slightly improve accuracy, simpler models remain highly competitive and offer better interpretability.

Problem Statement & Motivation

The goal of this project is to predict whether a patient has heart disease based on their medical attributes. This is a **binary classification problem**, where the model outputs either the presence or absence of heart disease.

This problem is important because early detection of heart disease can significantly improve patient outcomes. Machine learning models can assist doctors by identifying high-risk patients more quickly and accurately, supporting better decision-making in healthcare environments.

Dataset Description

The dataset used in this project comes from Kaggle and contains **1,025 patient records** with **13 features** and 1 target variable.

- **Features include:** age, sex, chest pain type, cholesterol level, blood pressure, etc.
- **Target variable:**
 - 1 = presence of heart disease
 - 0 = no heart disease

Data Quality:

- No missing values were found in the dataset
- The classes are relatively balanced (526 positive, 499 negative cases)

Preprocessing:

- Features were separated from the target variable
 - Data was split into **80% training and 20% testing**
 - Feature scaling was applied using **StandardScaler** to normalize input values
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Proposed Methodology

To evaluate different approaches, three machine learning models were implemented:

Baseline Model

- **Logistic Regression**
- Chosen because it is simple, fast, and highly interpretable

Additional Models

- **Decision Tree**
 - Captures non-linear relationships
 - Provides visual interpretability through tree structure
- **Neural Network (MLPClassifier)**
 - Introduced to test a more complex, non-linear learning method
 - Includes hidden layers (64 and 32 neurons)

Justification

These models were selected to compare:

- **Simple vs complex models**
 - **Interpretability vs performance**
 - How model complexity impacts results on a relatively small dataset
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Evaluation Plan

To assess model performance, I used multiple evaluation methods:

- **Accuracy**
- **Precision**
- **Recall**
- **F1 Score**

- **ROC-AUC Score**

Validation Strategy:

- **80/20 train-test split** with stratification
- **5-fold cross-validation** to measure consistency and reliability

Visualizations Used:

- Confusion Matrices
 - ROC Curves
 - Feature Importance (Decision Tree)
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Results and Analysis

Model Performance

- Logistic Regression: ~81% accuracy
- Decision Tree: ~84% accuracy
- Neural Network: ~84.4% accuracy

Although the Neural Network achieved the highest accuracy, the improvement over simpler models was minimal (~3.4%).

Key Observations

- Logistic Regression achieved the **highest ROC-AUC**, indicating strong overall classification ability
- Decision Tree showed **high stability** with the lowest variation during cross-validation
- Neural Network initially overfit the data (100% accuracy), but this was corrected using:
 - Early stopping
 - Regularization

Interpretation

The results suggest that for smaller, structured datasets:

- **Simple models can perform nearly as well as complex ones**
- The added complexity of Neural Networks may not always be justified

Feature Importance

- Chest pain type was the most influential feature (~43%)
 - Several features contributed very little, suggesting possible dimensionality reduction
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Limitations

One limitation of this project is the relatively small dataset size, which may restrict the ability of more complex models (like Neural Networks) to generalize effectively. Additionally, the dataset may not fully represent real-world patient diversity.

Conclusion

Machine learning models can effectively predict heart disease using patient data, achieving accuracy levels above 80%. While Neural Networks provided the highest performance, the improvement over simpler models was minimal.

Logistic Regression, despite being the simplest model, proved to be highly competitive and offered the advantage of interpretability — a critical factor in medical decision-making. This suggests that in real-world healthcare applications, simpler and more explainable models may often be preferred over more complex ones.